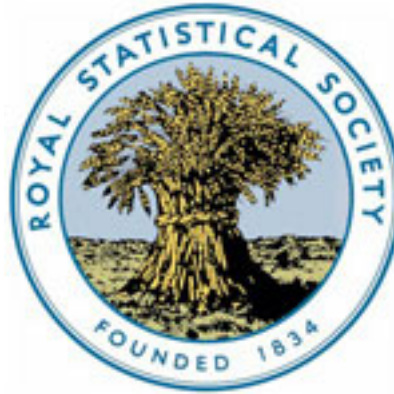


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Self-starting cusum charts for location and scale

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Abstract. In some quality control problems, it is not known what the exact process mean and standard deviation are under control but it is desired to determine whether there have been drifts from the conditions obtained at the process start-up. This situation is not well-covered by standard cumulative sum procedures, which generally assume known process parameters. This paper uses the running mean and standard deviation of all observations made on the process since start-up as substitutes for the unknown true values of the process mean and standard deviation. Using some theoretical properties of independence of residuals, two pairs of cusums are set up: one testing for constancy of location of the process, and the other for constancy of the spread. While the process is under control, both these cusum pairs are of approximately normal $N(0, 1)$ quantities (and therefore are well understood), but if the location, the spread or both change, then non-centrality is introduced into one or both of the location and scale cusum pairs, and it drifts out-of-control. It is shown that the procedure performs well in detecting changes in the process, even in comparison with the often utopian situation in which the process mean and variance are known exactly prior to the start of the cusum.

1 Introduction

Since their original proposal and discussion in, for example, Woodward & Goldsmith (1964), cumulative sum charts have become recognised as powerful quality control tools, especially for detecting relatively small departures from specification. Discussion of cusums can be found in many general texts on quality control, for example, Wetherill (1977) and Murdoch (1982), as well as Johnson & Leone (1977) and the series of papers by Johnson & Leone (1962) and a thorough, up-to-date and readable account of current thinking can be found in Lucas (1985). British Standard 5703 discusses the construction of cusum control charts in detail, and provides useful information on the choice of the levels of the decision limits.

The power of cusums for detection of even quite small departures from the supposed process mean carried the implication that this mean must be specified precisely if large numbers of spurious signals are to be avoided. While it has usually been assumed in discussion of cusums that the process specification is known exactly, there are many circumstances in which it is to some extent uncertain, and under these conditions the assumption of known mean and variance is inappropriate. A case of particular interest to the author arises in the control of assaying in a chemical laboratory for bias and precision. A good way of doing this is (Mandel 1964) by using reference materials. A reference material is a batch of the same sort of material as is assayed in the laboratory (obtained for example from a previous consignment), portions of which are assayed regularly along with the production material. A departure of the assays of the reference material from their previous mean or standard deviation indicates a change in bias or precision. Note that the control required is simply for any change in the values obtained for the reference material. It is immaterial what the true assay of the reference material (the ideal target process mean) actually is, and the quality control system should not be predicated on the assumption of an exact knowledge of the process mean. This indeterminacy presents

a problem for the direct implementation of standard cusum procedures, which assume known process mean and variance.

A common way of addressing this difficulty is by having portions of the reference material assayed repeatedly (e.g. Murdoch 1982 suggests 25 times and Nelson 1982 one or two dozen) as a separate start-up exercise. The mean and standard deviation so obtained are taken to be the true values, departure from which indicates a drift in the laboratory. This procedure is however undesirable on two counts: first, it may prove extremely difficult to carry out the special start-up assays under the same conditions as the subsequent production ones (as is required if the standard deviation in particular is to be realistic); and second, however many assays are performed in the start-up, there will remain some residual random variability in the mean and standard deviation estimated from them. This random variability will then become a systematic bias in the subsequent cusums and may lead to a wrong attempt to maintain the process at the level observed in the start-up exercise and not the true level.

In this paper, an alternative method is proposed. This alternative uses the actual sequence of observations on the process adaptively. It updates its estimates of the process mean and standard deviation at the same time as checking them with the cusum. It thus uses only production-condition measurements which has two desirable consequences. One is that the need for a special start-up exercise (and the associated effort and cost) is avoided. The other and more valuable (particularly as regards the process standard deviation) is that it ensures that the control applied to the observations is against a realistic and relevant standard actually attained under production conditions. The adaptive operation also ensures that the control incorporates more and more data into its estimate of the process mean and standard deviation so that the performance of the cusum improves steadily as time goes by.

2 Theoretical Basis

Assume that the process produces measurements X which, under control, are independent identically distributed random variables from a normal $N(\mu, \sigma^2)$ distribution.

Departures from control may take the form of shifts in μ or σ . These shifts will be thought of as starting at some instant, and persisting for some time. We assume that μ and σ are not known exactly *a priori*.

Let $\bar{X}_j$ be the mean of the first j of the X_j and S_j the sum of squares of deviations of these first j observations from $\bar{X}_j$. The statistics $\bar{X}_j$ and S_j can easily be obtained sequentially by using the following recursions:

$$\begin{aligned}\bar{X}_j &= \bar{X}_{j-1} + (X_j - \bar{X}_{j-1})/j \\ S_j &= S_{j-1} + (j-1)(\bar{X}_j - \bar{X}_{j-1})^2/j.\end{aligned}\tag{1}$$

Define

$$T_j = a_j(X_j - \bar{X}_{j-1})/\sqrt{[S_{j-1}/(j-2)]},\tag{2}$$

where

$$a_j = \sqrt{[(j-1)/j]}.$$

The studentised residual T_j is an intuitively very attractive test statistic for the compatibility of X_j with its predecessors, since it is just the Student's t test for testing the compatibility of the two 'samples' X_j and $\{X_1, \dots, X_{j-1}\}$. It thus makes the fullest possible use of the information at hand about the distribution of the observations prior to X_j . The studentised residuals, viewed as a sequence, also have helpful theoretical properties since, as has been shown previously, for example in Hawkins

(1969), the successive T_j are statistically independent, and T_j follows the well-understood t distribution with $j-2$ degrees of freedom.

While this statistical independence is very valuable in making it possible to control the run lengths of the cusums to be defined below, the t distribution (particularly in view of the number of degrees of freedom varying from one observation to the next) is not as convenient as the standard normal distribution would be. For this reason, we make use of one further theoretical result to produce a very nearly normal variable for cumulative summing:—

Define

$$U_j = \left(\frac{8\nu+1}{8\nu+3} \right) [\nu \log_e(1 + T_j^2/\nu)] \text{ where } \nu = n-2,$$

then U_j is approximately distributed as normal $N(0, 1)$. This approximation is due to Wallace (1959), and its accuracy for even a single degree of freedom ν has been reported by Peizer & Pratt (1968).

The interpretation of the derived quantities U_j is not immediately clear, and merits some comment. It is a function of the studentized deviation $(X_j - \bar{X}_{j-1})/s_{j-1}$ (where s_{j-1} represents the sample standard deviation of the first $j-1$ observations), and as $j \rightarrow \infty$, $U_j \rightarrow (X_j - \mu)/\sigma$. We may thus think of U_j as essentially just a studentised residual, restandardised to a $N(0, 1)$ distribution.

These preliminaries set the scene. The sequence U_j consists of independent approximately $N(0, 1)$ random variables, each of which measures the deviation of the corresponding X from its predecessors, using the fully efficient mean and standard deviation of the whole historical sequence. Our proposal is the following: as observations X_j are obtained, compute the quantities defined above. Maintain a cusum of the U_j to check for departures of the mean of U_j from zero. As we shall show below, any such departure indicates that the mean of the X_j has changed from its previous level.

For purposes of controlling the variability of the process, we make use of the scale cusum defined in Hawkins (1981). This is a cusum of the quantities

$$V_j = \sqrt{|U_j| - 0.822}/0.349. \quad (3)$$

The V_j are also very close to $N(0, 1)$ while the process is under control. They are not greatly affected by changes in the mean of the X_j , but a change in the standard deviation of the X_j leads to a change in their mean, and so a shift in level (as detected by a cusum) of the V_j is indicative of a change in the standard deviation of the X_j . Note for purposes of theoretical development that since the sequence V_j is a term by term transformation of the U_j , the V_j are also mutually independent.

Readers familiar with the cusum literature will recall that the cusum may be constructed in either of two mathematically equivalent ways—a ‘classical’ form, in which formal testing is done using a V-mask, and a ‘decision interval’ form, in which the action limits are horizontal lines. The V-mask variant is the better for visual display of the sequence, but is harder to use for formal inference—in particular it is not easy to see by inspection where different V masks would give rise to a signal. The decision interval form is less flexible in that it does not permit the use of different decision intervals on the same plot, but is easier to use for formal inference on whether changes have occurred since its decision limits are horizontal lines. While both forms have their place, we will concentrate for the present on the decision interval form and its derived decision rule.

In this form, we maintain altogether four cusums—for upward and for downward shifts in location; and for upward and for downward shifts in scale. Initially, we set up the four starting values:—

$$L_0^+ = L_0^- = S_0^+ = S_0^- = 0. \quad (4)$$

and choose two parameters k and h , values for which are discussed below. As values X_j become available, we update the cusums:

$$\begin{aligned} L_j^+ &= \max(0, L_{j-1}^+ + U_j - k) \\ L_j^- &= \min(0, L_{j-1}^- + U_j - k) \\ S_j^+ &= \max(0, S_{j-1}^+ + V_j - k) \\ S_j^- &= \min(0, S_{j-1}^- + V_j - k). \end{aligned} \quad (5)$$

The two L processes check for constancy of location (process mean), and the two S for constancy of scale, or standard deviation. Signals are given if any of them exceeds h in absolute value, and these signals interpreted as follows:—

- On L^+ :—the mean of the X_j has increased;
- on L^- :—the mean of the X_j has decreased;
- on S^+ :—the standard deviation of the X_j has increased;
- on S^- :—the standard deviation of the X_j has decreased.

This description of the operation of the cusum shows the meaning of the parameters h and k :— k is an ‘allowance per observation’—the cusum is updated not by the full deviation of U_j or V_j from specification, but only by the amount of deviation in excess of the allowance k . The parameter h is a decision limit—once a cusum exceeds h an out-of-control is signalled.

The parameters h and k are chosen by first deciding for what shift in mean one desires the best possible performance. Expressing this difference in as a multiple of the standard deviation and halving it gives the required value of k . The second part of the exercise then consists of choosing a suitable h . This is most easily done by looking up tables or charts of the average run length for different h values and the selected k , and choosing an h value that gives an acceptably high ARL under control. Such charts and tables may be found in, for example, Lucas (1976), and in part 2 of British Standard 5703.

While the process is in control, each of these cusums will characteristically ‘bump along’ on the axis at the value zero, then make an excursion away from zero. These excursions away from zero terminate, either in the cusum going back to zero (when the cycle of behaviour repeats) or in exceeding the value h and so giving an out-of-control signal. These typical excursions away from zero interspersing strings of zero values can be seen in Figs. 1 and 2. In diagnosing an out-of-control signal, one studies the excursion away from zero which ended in the signal. The first non-zero value in this excursion is the instant at which the process is diagnosed as having gone out of control, a very important feature of the diagnosis of the out-of-control.

All four of the cusums may be plotted on the same paper using different plotting symbols for the location and scale cusums. As L^+ and S^+ occupy the positive half of the plot, and L^- and S^- the negative this does not lead to the ‘chart clutter’ one might expect. In addition, with a suitable choice of scale the U_j can be superimposed on the cusum charts and used as a Shewhart chart for location and spread (as in Westgard *et al.*, 1977). The use of U_j to control for scale is an alternative to the proposal by Nelson (1982) to use the absolute value of successive differences, and has the advantage of using what is a better estimator of location while the process is under control.

Let us now look, initially rather informally, at the behaviour of the cusums proposed when there is a departure from the previous distribution of the data. We will assume that the distribution remains normal, but that the parameters make a single shift to new levels at which they remain. If the mean is shifted upward by an amount δ , say, from μ to $\mu + \delta$, then the mean of the numerator $a_j(X_j - \bar{X}_{j-1})$ of T_j is displaced from zero to $a_j\delta$, and so T_j becomes a non-central t variate with non-centrality $a_j\delta$. In turn, U_j is displaced from zero mean to a positive mean, and so the location cusum will have an expected upward displacement. If this first point is not large enough to give a

signal on the location cusum, then after the updating, the mean $\bar{X}_j$ will increase in expectation from μ to $\mu + \delta/j$. Despite this expected increase in $\bar{X}_j$, when the difference $X_{j+1} - \bar{X}_j$ is formed at the next stage of the cusum, its mean will still be substantially positive and so contribute a further upward expected movement to the cusum. It is easy to see, extending this argument, that if there is a single shift of mean from μ to $\mu + \delta$, then no matter how many further terms are added to the cusum, each will have a strictly positive expected deviation. Thus, even though the running mean tracks toward the new mean of the data, it always lags behind it, and the cusum continues to drift away from the axis, and this drift tends to lead to signals on the cusum chart.

The weight given to the new data relative to the history obviously depends on the length of the history. When the history is long, the historical information predominates, and the mean tracks very slowly toward its new level. This implies that the upward drift in the cusum is sharp and the detection of the shift in mean swift. When the history length is short, then the mean tracks more quickly toward the new level, and in consequence of this, the upward drift in the cusum, while still present, is more muted, and the detection of the change is slower and less certain. We may quantify this effect of the length of previous history under control by computing, for different j , what submultiple of δ feeds through into the upward drift of the cusum for the first few points after a shift in location.

Table 1. Proportion of mean shift reflected in non-centrality of first few points

Point	History length		
	25	50	100
1	0.943	0.971	0.985
2	0.909	0.952	0.976
3	0.877	0.934	0.966
4	0.747	0.917	0.957
5	0.819	0.901	0.948
6	0.793	0.885	0.939
7	0.769	0.869	0.930
8	0.746	0.855	0.922
10	0.704	0.826	0.905
20	0.549	0.709	0.830

Table 1 shows these values, showing for example that if the process runs under control for 50 points and then moves in mean by an amount δ , the successive T_j will have non-centralities of 0.971δ , 0.952δ , 0.934δ , 0.917δ , and so on. All of these are large submultiples of δ , and so imply that the self-starting cusum will respond almost as quickly to a shift in mean as it would if the process mean and standard deviation had been estimated from an infinitely long history. These non-centrality calculations suggest that unless the in-control learning period is very short, the non-centralities will remain at perhaps surprisingly large fractions of δ , and hence the cusum will perform quite comparably with the ideal case in which the process mean and standard deviation had been estimated from an 'infinitely long' history (say 100 or more) so that they were known without discernable error.

One primary effect of a shift in mean of the X_j is on the location cusum, but as a result of this displacement in mean of the numerator of T_j , the variance of U_j will in general also be affected, though to a much smaller extent. Thus while the primary effect of the shift in mean is on the location cusum, there will also be a much smaller and generally negligible effect on the scale cusum.

A shift in scale, by contrast, becomes a shift in the location of the V_j , and so will translate into an upward or downward drift on the scale cusum. Such a shift will show

its presence primarily in the scale cusum, but will also have a secondary effect on the location cusum. This secondary effect arises because the parameters k and h are actually in units of the true standard deviation of the data. Thus if the true standard deviation increases above the historical level, then the h and k values are being scaled by too low a standard deviation, and so the upward shift in the process standard deviation has the same effect on the location cusum as a proportional reduction in both the h and k mask parameters—namely a reduction in the ARL of that cusum. Thus the effect of a scale shift may appear in the location cusum, and not just in the scale cusum.

As with the location cusum, as long as the scale change remains unsignalled, the contaminated observations will feed into the running variance and increase it, but this variance will always lag behind the variance of the current observations so that the scale cusum continues to drift away from the horizontal.

The practical implication of these theoretical properties is: when only the location cusum gives a signal, it may be taken to indicate a location shift. When only the scale cusum gives a signal, it indicates a scale shift. When both cusums give a signal, this may indicate a scale shift only, and not necessarily a shift of both location and scale. In practice, this indeterminacy is easily resolved. When a signal is given, the first step is to establish when the process moved out of control. This is done by taking whichever of the four decision interval cusums L^+ , L^- , S^+ and S^- gave the signal, and looking up the values preceding the signal to find the instant at which the process moved out of control. One then finds the means and standard deviations of the observations preceding the change point; and of those from the change point up to the current point. (Efficient computational methods for doing this will be given under the heading ‘Implementation details’). From these ‘before and after’ calculations carried out when a signal has occurred, it is easy to decide to what extent the signal is due to location shifts and to scale shifts.

It may be noted that the first two observations do not give rise to T_j or U_j since it is only with the third observation that a non-trivial estimate of the standard deviation becomes available. Thus cusum points cannot be plotted for these two points, and the cusums start from the third observation of the sequence. On occasion, one of the cusums will immediately start an excursion away from the axis that ends in a signal. This situation will usually be diagnosed as a start-up problem affecting the first few observations of the sequence, and it may be felt advisable to restart the cusum with the first few points omitted.

3 Performance of the cusums

The preceding rather informal remarks about the power of the cusums will now be expanded with some average run length (ARL) results. The scheme was studied by simulating sequence of standard normal data. Random normals from a fixed distribu-

Table 2. ARL of the cusums following shifts

ARL for δ	K	History length in control							
		25	50		75		100		
		loc	scale	loc	scale	loc	scale	loc	scale
0	1	124	121	115	112	113	110	110	110
0.25	1	97	123	79	114	69	107	61	102
0.5	1	58	127	31	113	26	102	22	98
0.75	1	27	129	14	107	12	86	11	77
1	1	12	105	9	72	8	54	7	42
0	1.5	69	59	47	31	37	22	32	19
0	2	56	19	28	10	20	9	17	9

(Here, ‘loc’ refers to the location cusum and ‘scale’ to the scale.)

tion were generated for 25, 50, 75 and 100 time points, after which shifts were made in either the mean or variance. The process mean (standard deviation) was changed from 0 to δ (1 to K) respectively. The location and scale cusums were tested at each point for a departure from specification. The cusum parameters for both the location and scale cusums were $h=6$ and $k=0.25$, as used in Hawkins (1981), giving an ARL under control of about 100 on each of the cusums (some comments will be made in the discussion of an example below on this choice of h and k). The ARL of this cusum was estimated for each departure from specification in the simulation, and the results are given in Table 2.

The striking feature of this table is that the effect of the length of history while the process is under control, while substantial when going from a 25 point history to a 50 point, is not very marked at the longer lengths.

This observation has interesting implications, not only for the proposals made here, but also for the more common practice of using a special start-up series of observations. The latter approach may be considered to be very much the same as using the self-start cusums, but 'switching off' the updating of the mean and standard deviation at the end of the start-up series and continuing to use the mean and standard deviation as at that point. The table shows that the use of about 25 start-up observations (as seems to be standard practice) is too short a learning set, particularly as regards the process standard deviation. This is shown, for example, by the fact that if 50 observations rather than 25 are used for the calibration, then the response to a doubling of the process variance is twice as fast. Now if the start-up values were obtained by a special study, it might be hard to justify the doubling of the start-up study just for the sake of the improvement in power of the cusum; when however the data are the very data produced by the process and involve no extra measurement expense or delay, this objection cannot arise.

Thus the self-starting cusums produce superior performance to those obtained with some 25 special start-up values while not involving any measurements beyond those produced anyway. The implications of this will be sketched in the Conclusion. We would therefore argue that self-start cusums should always be used in preference to plugging the mean and observations of a start-up sample into the formulae assuming known parameters. If the start-up exercise is carried out, its observations can be used as the first production data. This has the advantage that the process mean and standard deviation can be determined more precisely (and hence the ARL properties of the cusums improved) with no additional cost of experimental observations. The only situation in which this could be inferior to the conventional use of a start-up sample would be where the production observations obtained right after the start-up sample were already out of control, but by an amount both large enough to seriously contaminate the running mean and standard deviation, and yet too small for detection by the cusums. Only under these rather specific circumstances, is the use of self-start cusums not clearly preferable to those obtained in the more traditional way.

The other useful implication in Table 2 flows from the small improvement as one goes to the longer history lengths. This means that the value of exact information on the process mean and standard deviation is quite small once one has passed the 50 observation mark.

As predicted by the theoretical considerations of Section 2, the table also shows a good degree of separability in the performance of the cusums. As expected, a location shift will trigger the location cusum, but has hardly any effect on the scale cusum unless the shift is very substantial, and while a shift in scale affects the scale cusum the more markedly, it also has some tendency to trigger the location cusum.

4 Implementation detail

As mentioned, the cusums are most conveniently calculated and used in the decision

interval form. Not only does this confer computational advantages, but it also leads to the use of horizontal (and therefore easily understood) decision limits, and has the property which is liked by users that after action is taken and the process brought back under control, the cusum drifts back towards the axis rather than running on horizontally.

In an actual routine implementation of these proposals, we are using a combined Shewhart/cusum chart. Both the positive and negative decision intervals are plotted on the same paper, with the positive one above the axis, and the negative one below. The Shewhart chart is superimposed in a different colour. The scales of the Shewhart and cusums are chosen so that the same horizontal line serves as the decision limit of each. A Shewhart signal is interpreted as indicative, not of bias, but of a likely error of execution on a single sample, and this sample is then remeasured. A cusum signal calls for a general review of methodology, and other actions to remedy the effects of bias on materials handled during the currency of the excursion away from the axis which led to the signal. These combined charts are easily understood by the line staff whose performance is monitored by them.

Another implementation detail for our application of the methodology perhaps deserves mention. As the process generating our data is outlier-prone (generates occasional valid observations far out into the tails of the distribution even when in control), we have found it advisable to robustify the cusum. There are many possible ways of achieving such robustification—for example Lucas & Crosier (1982) use the rejection of isolated outliers—but we have settled in preference on an ‘accommodation’ approach (Hawkins, 1980) using winsorisation. In this, any U_j value exceeds a preset cutoff (usually 3) is edited down to equal the cutoff and the edited value used in the cusum. As a consequence of this, the cusum is protected from signals generated solely by isolated outliers. This is appropriate to applications such as this, where the primary role of the control is detecting small shifts in mean—the outliers are diagnosed using the Shewhart part of the control, while still being allowed to make a limited contribution to the cusum. These parts of the implementation are rather specific to our application, and robustification may be eschewed or done in other ways if the basic process is not outlier-prone.

When a signal is given by one or the other cusum, one may wish to obtain the mean and variance, say $\bar{X}$ and s^2 , of the sub-sequence of observations X_{k+1} to X_j during which the process is diagnosed as being out of control. It is not necessary to compute these de novo as they can be derived from the $\bar{X}_k$ and S_k values using the following analysis of variance identities:—

$$\bar{X} = (j\bar{X}_j - k\bar{X}_k) / (j - k), \quad (6)$$

$$(j - k - 1)s^2 = [S_j - S_k - kj(\bar{X}_j - \bar{X}_k)^2 / (j - k)]. \quad (7)$$

For these identities to be useful, one must retain the sequence of $\bar{X}_k$ and S_k . This is perhaps most easily done by printing them, or equivalent quantities on the cusum alongside the points.

Since data are never known to be exactly normally distributed, another question that arises is the effect of departure from the assumption of normality of the sequence X_j . The fact that T_j is studentised using data from the same distribution reduces the problem of non-normality of the X_j to the problem of the applicability of the Student t test to non-normal data. As shown, for example in Tan (1982) and Yuen & Murthy (1974), under these circumstances the t test is applicable, but slightly conservative if the underlying distribution is heavy tailed (as is often the case if normality is not present). Thus the method can also be expected to perform well with non-normal heavy-tailed data, though the run lengths will tend to be somewhat longer than is the case with exactly normal data.

5 Example

The cusums will be illustrated by an example of two laboratories carrying out routine indirect (instrumental) assays for precious metals of batches of a feedstock. It is difficult to carry out such routine assays to the level of accuracy and precision required, and as a control measure, a sample of a standard reference material is assayed along with each batch of unknowns. The nature and function of the reference material may deserve some comment. Some time ago, one batch of the test material was selected, and a portion extracted from it. This portion was mixed thoroughly and split down into a large number of assay portions, which are kept on hand in the laboratory as a reference material. Whenever a batch of new material is to be assayed, one of the portions of reference material is taken, and assayed alongside the new unknown material. The rationale for this is that, if any aspect of the assay changes and gives rise to a bias (or a shift in the bias) of the assays, then this can be detected in that the assay of the reference material will change from the values obtained previously. Such otherwise undetected changes in the assay bias can in fact occur very easily as a result of, for example, drift in the equipment; an error in mixing internal standards; personnel changes and so on. The use of the reference material allows such bias to be detected in the change in the level and/or variability of its assays. Note that the actual grade (concentration of precious metal) of the reference material is completely immaterial for this control; all that is needed to control for possible new biases is to detect changes in the assay over time, and for this the self-starting cusums are particularly appropriate.

A question arises naturally of what cusum parameters h and k should be chosen for the best practical statistical bias control in such an application of a reference material. The considerations that arise here are (i) that a bias much over one standard deviation should never arise in the first place if normal care and attention are used by the assayers, and if it were to arise, should become apparent to them very soon; (ii) while no bias of whatever magnitude is acceptable, the harm done by a bias goes up sharply with its magnitude, and with the length of time for which it goes undetected; and (iii) a bias below about one quarter of a standard deviation would take an inordinately long time to detect even using cusums, by which time it would be of largely historical interest. This suggests that the region in which good diagnostics are both needed and likely to be effective is about one half a standard deviation. Tuning the cusum for optimal detection of a bias of 0.5 standard deviations implies a value of $k=0.25$. The choice of $h=6$ follows thereafter from the decision that an ARL under control of about 100 would be right in terms of the amount of effort expended on investigating false alarms.

A set of real data from each laboratory is given in coded form in Table 3. The coding—necessary to preserve the confidentiality of the data—was carried out by subtracting from the actual assays the mean of the whole set, and then dividing by the standard deviation.

The data have been analysed by self starting cusums using the decision interval parameters $k=0.25$, $h=6$. The resultant decision interval location and scale cusums are shown as Figs 1 and 3 with the two location cusums plotted as solid lines and the scale as dashed ones. Conventional cusums for location and scale are also shown as Figs 2 and 4. The printouts corresponding to the Figures are given as Tables 4 and 5.

The cusums for laboratory 1 show that the process variability has not gone outside the decision limits of the cusum at any stage of operation, and so appears to have remained stable. The location cusum on the other hand, has gone outside limits, with an upward excursion that crosses the line $h=6$. Either by reading off the chart or by going to the printout, we see that the 30th observation gave an L^+ value larger than 6, while the 16th was the first point in that excursion whose L^+ was nonzero. We thus conclude that the mean assay increased significantly at the 16th observation in the series, so that after an initial 15 observations at one level, the process mean has shifted to another level.

At this stage, it may be of interest to compute the mean and standard deviations of

Table 3. Sequence of data from the two laboratories

Laboratory 1									
0.82	0.40	-2.02	-0.02	-2.18	-0.64	-0.39	-0.51	1.17	0.49
-1.77	-0.64	-2.30	-1.55	-0.90	0.03	0.50	0.60	-0.65	0.19
-0.38	-0.72	-0.21	-0.50	0.95	1.59	0.68	-0.34	0.30	2.23
-0.75	1.39	1.01	-0.80	0.15	1.37	-1.39	0.86	0.64	-0.21
0.51	-0.21	0.51	0.12	-0.33	1.01	-1.34	1.01	-0.04	1.67
1.26	-0.01	0.06	-0.82	0.12					
Laboratory 2									
0.30	0.18	-1.61	-0.38	0.18	-0.38	0.45	-0.53	-0.65	-0.29
1.50	-0.74	-1.10	0.33	-0.26	-0.20	-0.47	-1.22	-0.47	-0.68
-1.07	-1.37	1.44	-0.50	-0.83	0.15	0.84	0.54	2.51	-0.03
0.36	-0.53	1.68	1.35	1.18	2.24	-0.23	2.24	-0.86	0.15
0.90	0.69	-0.47	1.32	-0.38	0.69	1.80	-0.53	-1.10	-1.04
0.45	-1.04	0.90	0.92	1.26	0.00	0.48	-0.47	-1.64	1.08
0.89	0.45	-0.11	-1.22	-2.21	0.06	-0.26			

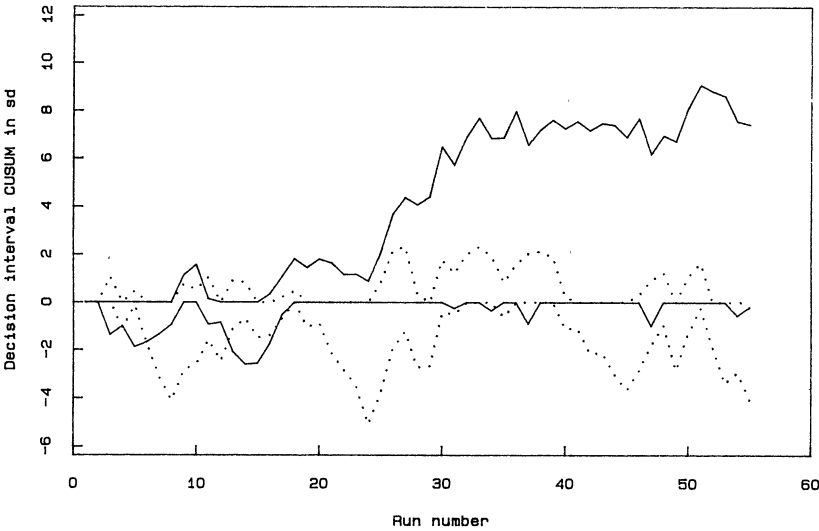


Fig. 1. Decision interval cusums for Laboratory 1.

the assays within each of these two segments. This is quite easily done using the running means and standard deviations from the printout, giving the following results:

Observation 15: $\bar{X}_{15} = -0.670$; $SD = 1.077$; $S_{15} = 17.399$

Observation 30: $\bar{X}_{30} = -0.192$; $SD = 1.066$; $S_{30} = 34.091$

Mean and variance of first segment = -0.670 and 1.243 ,
from which using equations (6) and (7)

Mean of new segment = $(-30 \times 0.192 + 14 \times 0.670) / 15 = 0.286$

Variance of new segment = $(34.091 - 17.399 - 6.855) / 14 = 0.703$.

It is striking that the change in the mean from the first segment to the second is less than the process standard deviation of (on the transformed scale) 1, and this indicates the sensitivity of cusum to process shifts not detectable by visual inspection or by non-cumulative methods of quality control.

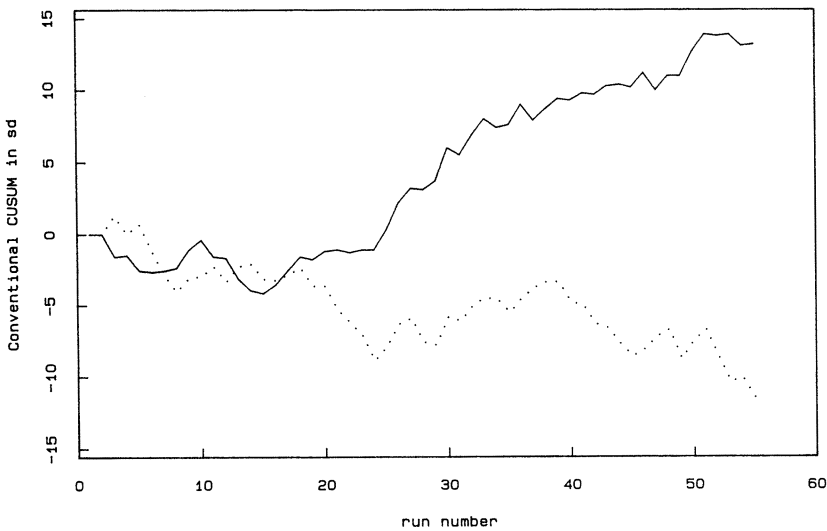


Fig. 2. Conventional cusums for Laboratory 1.

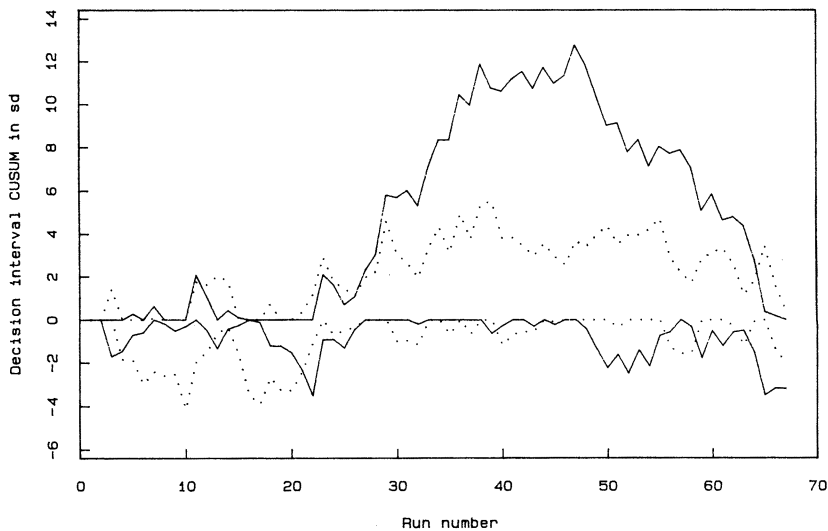


Fig. 3. Decision interval cusums for Laboratory 2.

Once the shift in location has been discovered, the correct course of action is a chemical investigation to locate any procedural or other changes that may have occurred at the indicated time of the change of regime, and any improper aspects of these in the first regime; the second; or both can then be rectified. In addition to this conventional response to a quality control signal, a further operation that may be considered appropriate is a post hoc adjustment of the assays of production material carried out while the process was in the new bias regime. If it is assumed (not unreasonably) that the unknowns will be biased by the same amount as the reference materials, then the subtraction of $(0.670 + 0.286) = 0.956$ from all assays carried out in the second regime will make them comparable to assays carried out in the first.

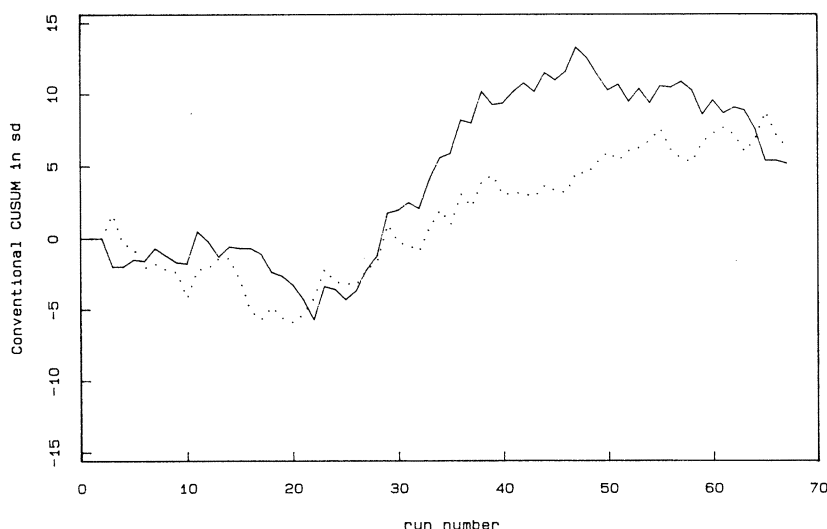


Fig. 4. Conventional cusums for Laboratory 2.

The cusums of the second laboratory are given in Figs 3 and 4 and Table 5. Here again, there has been no significant departure on the scale cusum, but an even sharper shift has occurred on the location cusum. The first signal occurs at point 31, and indicates a change of location after 22. The means and variances of the two segments identified at this point are as follows:—

Observation 22: $\bar{X}_{22} = -0.387$; $SD = 0.688$; $S_{22} = 10.414$

Observation 31: $\bar{X}_{31} = -0.135$; $SD = 0.858$; $S_{31} = 22.821$

Mean and variance of first segment are -0.387 and 0.496 ,
from which again (6) and (7) yield

Mean of new segment $= (-31 \cdot 0.135 + 22 \cdot 0.387) / 9 = 0.481$,

Variance of new segment $= (22 \cdot 876 - 10 \cdot 414 - 4 \cdot 846) / 8 = 0.945$.

This shift (whose high statistical significance was confirmed in the subsequent values of the cusum—see Fig. 3) was again less than one standard deviation in magnitude, and was not detected by Shewhart charts. The cusum detected it very quickly—after only 9 out-of-control values. This set of data is also interesting in that it contained an outlier—observation 29—which was winsorised down to three standard deviations by the program.

6 Other applications

While the technique set out here was originally motivated by a specific quality control problem, it has more general quality control uses, and has proved extremely useful in other problems of analysis of time series data. Taking the former first, we may draw a distinction in quality control between normative problems and descriptive ones. In a normative problem, we strive to attain some target level for the process. Unless this target is in fact well within reach of the process, cusums are not very suitable for this, as they continue to document the failure to maintain the target within the fine discrimination levels they make possible. In a descriptive problem, the target is obtained from the process capability as actually measured. For problems of this type

Table 4. Printout for Laboratory 1. X represents the successive data values, and U the $N(0,1)$ studentised deviation from the running mean.

SELF STARTING	CUSUMS	FOR LOCATION AND	SCALE	CONVENTIONAL	CUSUMS	DECISION	INTERVAL	CUSUMS		
NO	X	U	RUNNING MEAN	S.D.	LOCATION	SCALE	LP	LM	SP	SM
1.	.821	0.00	.821	0.000	0.0	0.0	0.00	0.00	0.00	0.00
2.	.397	0.00	.609	.212	0.0	0.0	0.00	0.00	0.00	0.00
3.	-2.021	-1.63	-.268	1.252	-1.6	1.3	0.00	-1.38	1.05	0.00
4.	-.018	.13	-.205	1.090	-1.5	-.0	0.00	-1.00	0.00	-1.09
5.	-2.182	-1.14	-.601	1.255	-2.6	-.7	0.00	-1.89	.46	-.13
6.	-.640	-.02	-.607	1.146	-2.7	-1.2	0.00	-1.67	0.00	-1.79
7.	-.513	.07	-.594	1.061	-2.6	-2.9	0.00	-1.35	0.00	-3.16
8.	-.386	.16	-.568	.995	-2.4	-4.1	0.00	-.94	0.00	-4.11
9.	1.167	1.38	-.375	1.085	-1.1	-3.1	1.13	0.00	.76	-2.85
10.	.489	.68	-.289	1.062	-.4	-3.0	1.56	0.00	.52	-2.59
11.	-1.769	-1.18	-.423	1.098	-1.6	-2.3	.13	-.93	1.02	-1.59
12.	-.639	-.18	-.441	1.053	-1.7	-3.4	0.00	-.85	0.00	-2.49
13.	-2.297	-1.50	-.584	1.126	-3.2	-2.3	0.00	-2.10	.91	-1.09
14.	-1.549	-.77	-.653	1.113	-4.0	-2.1	0.00	-2.62	.81	-.68
15.	-.904	-.21	-.670	1.077	-4.2	-3.2	0.00	-2.58	0.00	-1.49
16.	.028	.59	-.626	1.057	-3.6	-3.3	.34	-1.74	0.00	-1.39
17.	.500	.97	-.560	1.059	-2.6	-2.9	1.06	-.52	.21	-.67
18.	.604	1.00	-.495	1.063	-1.6	-2.4	1.81	0.00	.48	0.00
19.	-.651	-.14	-.503	1.035	-1.8	-3.7	1.43	0.00	0.00	-1.04
20.	.189	.62	-.469	1.020	-1.2	-3.7	1.80	0.00	0.00	-.89
21.	-.375	.09	-.464	.996	-1.1	-5.3	1.63	0.00	0.00	-2.15
22.	-.719	-.24	-.476	.974	-1.3	-6.2	1.14	0.00	0.00	-2.85
23.	-.215	.25	-.465	.954	-1.1	-7.1	1.15	0.00	0.00	-3.52
24.	-.501	-.04	-.466	.934	-1.1	-8.9	.86	0.00	0.00	-5.08
25.	.948	1.40	-.410	.956	.3	-7.9	2.02	0.00	.79	-3.79
26.	1.593	1.91	-.333	1.014	2.2	-6.3	3.68	0.00	2.15	-1.93
27.	.685	.95	-.295	1.013	3.2	-5.9	4.38	0.00	2.33	-1.24
28.	-.341	-.04	-.296	.995	3.1	-7.6	4.08	0.00	.32	-2.76
29.	.304	.58	-.276	.984	3.7	-7.8	4.41	0.00	0.00	-2.69
30.	2.226	2.32	-.192	1.066	6.0	-5.8	6.48	0.00	1.76	-.43
31.	-.754	-.50	-.211	1.054	5.5	-6.1	5.72	-.25	1.19	-.50
32.	1.386	1.43	-.161	1.074	6.9	-5.0	6.90	0.00	2.01	0.00
33.	1.006	1.04	-.125	1.076	8.0	-4.5	7.69	0.00	2.32	0.00
34.	-.800	-.60	-.145	1.066	7.4	-4.6	6.84	-.35	1.93	0.00
35.	.154	.27	-.137	1.052	7.6	-5.5	6.86	0.00	.82	-.62
36.	1.374	1.37	-.095	1.067	9.0	-4.5	7.97	0.00	1.56	0.00
37.	-1.388	-1.16	-.130	1.073	7.9	-3.8	6.56	-.91	2.04	0.00
38.	.856	.88	-.104	1.070	8.7	-3.4	7.20	0.00	2.13	0.00
39.	.639	.67	-.085	1.063	9.4	-3.4	7.61	0.00	1.87	0.00
40.	-.214	-.12	-.088	1.050	9.3	-4.8	7.25	0.00	.25	-1.12
41.	.511	.55	-.073	1.041	9.8	-5.0	7.55	0.00	0.00	-1.10
42.	-.214	-.13	-.077	1.029	9.7	-6.3	7.17	0.00	0.00	-2.17
43.	.511	.55	-.063	1.021	10.3	-6.6	7.47	0.00	0.00	-2.14
44.	.120	.17	-.059	1.009	10.4	-7.7	7.39	0.00	0.00	-3.05
45.	-.329	-.26	-.065	.999	10.2	-8.6	6.88	-.01	0.00	-3.69
46.	1.006	1.04	-.041	1.000	11.2	-8.1	7.67	0.00	.31	-2.88
47.	-1.342	-1.25	-.069	1.007	10.0	-7.2	6.17	-1.00	.91	-1.78
48.	1.007	1.03	-.047	1.008	11.0	-6.7	6.95	0.00	1.22	-.97
49.	-.042	.01	-.047	.998	11.0	-8.8	6.71	0.00	0.00	-2.87
50.	1.674	1.66	-.012	1.017	12.7	-7.5	8.11	0.00	1.08	-1.29
51.	1.259	1.21	.013	1.022	13.9	-6.7	9.07	0.00	1.63	-.24
52.	-.007	-.02	.012	1.012	13.8	-8.6	8.80	0.00	0.00	-1.95
53.	.062	.05	.013	1.003	13.9	-10.4	8.60	0.00	0.00	-3.43
54.	-.823	-.81	-.002	1.000	13.1	-10.1	7.54	-.56	0.00	-2.96
55.	.120	.12	-.000	.991	13.2	-11.5	7.41	-.19	0.00	-4.07

we can see considerable scope for self-starting cusums. They can be incorporated in the process capability study, and simply continued with production data thus ensuring that all good data obtained on the system are incorporated into the running mean and standard deviation.

When departures from target are detected and corrected, the cusums can be back-tracked to use the last mean and standard deviation prior to the onset of the out-of-control, thereby ensuring that these running means do not contain significant contamination. The additional information contained in the ongoing data provides a significant enhancement to the performance of the cusum over that obtained from the use of the data from the initial process capability study alone.

Turning to the application in more general data analysis, when faced with a set of data in time (or other) order, it is often very informative to set up the self-starting cusum as a supplement to non-time-order methods such as stem and leaf plots. The cusum provides powerful diagnostics for variations in the level or variability of the sequence.

Similarly, application of the method to the recursive residuals of a set of regression data with some ordering provides an alternative, and in many ways preferable, analysis to that proposed by Brown *et al.* (1975), or by Galpin & Hawkins (1984).

Table 5. Printout for Laboratory 2. X represents the successive data values, and U the $N(0,1)$ studentised deviation from the running mean

SELF NO	STARTING X	CUSUMS U	FOR LOCATION AND SCALE		CONVENTIONAL LOCATION	CUSUMS SCALE	DECISION LP	INTERVAL LM	CUSUMS SP	CUSUMS SM
			MEAN	S.D.						
1.	.302	0.00	.302	0.000	0.0	0.0	0.00	0.00	0.00	0.00
2.	.185	0.00	.244	.059	0.0	0.0	0.00	0.00	0.00	0.00
3.	-1.609	-1.97	-.374	.874	-2.0	1.7	0.00	-1.72	1.42	0.00
4.	-.384	-.01	-.376	.757	-2.0	-.4	0.00	-1.48	0.00	-1.86
5.	.183	.52	-.264	.713	-1.5	-.7	.27	-.71	0.00	-1.91
6.	-.383	-.13	-.284	.653	-1.6	-2.1	0.00	-.59	0.00	-2.99
7.	.452	.87	-.179	.657	-.7	-1.8	.62	0.00	.07	-2.42
8.	-.533	-.44	-.223	.626	-1.2	-2.2	0.00	-.19	0.00	-2.61
9.	-.653	-.58	-.271	.605	-1.7	-2.4	0.00	-.52	0.00	-2.54
10.	-.294	-.03	-.273	.574	-1.8	-4.2	0.00	-.31	0.00	-4.13
11.	1.498	2.31	-.112	.748	.5	-2.2	2.06	0.00	1.75	-1.88
12.	-.742	-.74	-.165	.737	-.2	-2.1	1.07	-.49	1.61	-1.52
13.	-1.101	-1.11	-.237	.750	-1.3	-1.4	0.00	-1.35	2.02	-.61
14.	-.333	.68	-.196	.738	-.6	-1.4	.43	-.42	1.78	-.35
15.	-.264	-.08	-.201	.713	-.7	3.0	.10	-.25	.00	-1.62
16.	-.205	.01	-.201	.690	-.7	5.1	0.00	-.01	0.00	-3.52
17.	-.474	-.36	-.217	.673	-1.1	-5.7	0.00	-.12	0.00	-2.68
18.	-1.220	-1.34	-.273	.693	-2.4	-4.8	0.00	-1.22	.72	-2.68
19.	-.473	-.27	-.283	.676	-2.7	-5.6	0.00	-1.23	0.00	-3.30
20.	-.683	-.55	-.303	.665	-3.3	-5.9	0.00	-1.54	0.00	-3.28
21.	-1.071	-1.07	-.340	.669	-4.3	-5.3	0.00	-2.35	.36	-2.42
22.	-1.369	-1.41	-.387	.688	-5.7	-4.2	0.00	-3.52	1.15	-1.12
23.	1.438	2.34	-.307	.769	-3.4	-2.2	2.09	-.93	2.93	0.00
24.	-.503	-.24	-.315	.754	-3.6	-3.1	1.60	-.92	1.73	-.70
25.	-.833	-.65	-.336	.745	-4.3	-3.2	.70	-1.32	1.44	-.50
26.	.154	.62	-.317	.737	-3.7	-3.3	1.07	-.44	1.09	-.34
27.	.841	1.46	-.274	.755	-2.2	-2.2	2.29	0.00	1.95	0.00
28.	.542	1.02	-.245	.757	-1.2	-1.6	3.06	0.00	2.24	0.00
29.	2.514	3.00	-.156	.882	1.8	1.0	5.81	0.00	4.60	0.00
30.	-.025	.14	-.151	.867	2.0	-.3	5.70	0.00	3.07	-1.03
31.	-.363	.57	-.135	.858	2.5	-.5	6.02	0.00	2.63	-.97
32.	-.533	-.45	-.147	.847	2.1	-.9	5.32	-.20	1.93	-1.17
33.	1.677	2.00	-.092	.891	4.1	-.8	7.08	0.00	3.38	0.00
34.	1.348	1.53	-.050	.911	5.6	2.0	8.35	0.00	4.32	0.00
35.	.184	.25	-.043	.899	5.9	1.0	8.35	0.00	3.14	-.68
36.	2.245	2.36	-.021	.962	8.2	3.1	10.45	0.00	4.93	0.00
37.	-.234	-.26	.014	.950	8.0	2.2	9.95	-.01	3.77	-.66
38.	2.244	2.19	.072	1.003	10.2	4.0	11.89	0.00	5.41	0.00
39.	-.861	-.89	.048	1.001	9.3	4.4	10.75	-.64	5.52	0.00
40.	.153	.10	.051	.989	9.4	3.0	10.60	-.29	3.82	-1.19
41.	.901	.83	.072	.985	10.2	3.2	11.18	0.00	3.83	-.69
42.	.691	.61	.087	.978	10.8	3.1	11.53	0.00	3.45	-.56
43.	-.473	-.55	.074	.970	10.2	2.9	10.73	-.30	2.98	-.53
44.	1.319	1.24	.102	.977	11.5	3.7	11.72	0.00	3.56	0.00
45.	-.384	-.48	.091	.969	11.0	3.3	10.98	-.23	2.95	-.11
46.	.691	.60	.104	.962	11.6	3.2	11.33	0.00	2.56	0.00
47.	1.797	1.68	.140	.983	13.3	4.6	12.77	0.00	3.68	0.00
48.	-.533	-.67	.126	.977	12.6	4.5	11.85	-.42	3.41	0.00
49.	-1.101	-1.21	.101	.983	11.4	5.3	10.39	-1.38	3.96	0.00
50.	-1.041	-1.12	.078	.986	10.3	6.0	9.02	-2.25	4.39	0.00

7 Conclusion

The ideas discussed in this paper have two broad implications for quality control using cusums. The first relates to the situation modelled directly—that where the process mean and/or variance are not known *a priori* and have values that are not of primary importance. Under these circumstances, some modification to the conventional cusum methods is needed. It has been shown that the sequence of successive residuals of each observation from all its predecessors, studentised by the standard deviation of these predecessors, and having the necessary statistical properties of independence and known distribution, permits the setting up of location and scale cusums whose performance is almost as good as that obtained if the true parameter values were really known, and much superior to that obtained by getting the mean and standard deviation of a startup sample and using these values in place of the unknown true process mean and standard deviation. Self-starting cusums have two advantages over the more traditional approach of carrying out a special process capability study and using the mean and standard deviation obtained as the ‘true’ values—(i) they avoid the necessity and the implied cost in time and money of the separate study; as well as the possible difficulties that such studies may produce atypical values; and (i) they have longer ARL’s under control and shorter ARL’s out of control than such schemes because of the steady improvement obtained in the estimates of the process mean and standard deviation. They have the disadvantage of requiring considerably

more arduous arithmetic than 'known parameter' cusums, but this disadvantage only comes into serious consideration if the calculations are done by hand.

The other implication is in the more general context of cusum work. There are many applications currently treated as problems of control to known parameters which are actually much closer to the situation described here. In them, it is felt that the process is currently running at an acceptable level, and that the future control should test for departures from this level. The standard procedure is then to carry out a process capability study, obtaining an $\bar{X}$ and s^2 , and then to plug these into the 'known parameter' as if they were the true values under control. We would argue that self-start cusums should always be used in preference to this procedure. While this obviates the need for the special capability study, if such an exercise is carried out, its observations can be used as the first data to go into the self-starting cusum, which can continue with further production data. The use of the self-starting cusum has the advantages that the process mean and standard deviation become steadily better and better estimated as data accumulate, and this gives both shorter and better-controlled ARL's at no cost in additional observations.

Apart from these traditional quality control uses, the methodology has applications to general time series analysis, and is also useful for many problems as a routine data analysis tool.

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Note added in proof

The author's attention has recently been drawn to a body work done by C. P. Quesenberry since 1973 under the heading of the 'conditional probability integral transform' which also makes use of the distribution and independence of the successively studentised residuals. For example, in Quesenberry (1986) a test for outliers equivalent to our Shewhart procedure is given, and in Hester & Quesenberry (1984) recursive residuals are used as the basis of a test for monotonic heteroscedasticity. The latter report also provides useful references to earlier papers in the area. The author is grateful to Dr Quesenberry for providing a set of his reprints and the opportunity to reference this work.

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Appendix 1

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C      SUBROUTINE SSCUSM(X,N,AK,ISPAC)
C      FORTRAN IV SUBROUTINE FOR IMPLEMENTATION OF SELF-STARTING
C      CUSUMS.  PARAMETERS ARE AS FOLLOWS:-
C      X THE VECTOR OF DATA TO BE CUSUMMED
C      N THE NUMBER OF VALUES IN X.
C      AK IS THE K PARAMETER OF THE CUSUM V MASK IMPLICIT
C      IN THE DECISION INTERVALS.
C      ISPAC IS SOMETIMES USED IF VERY LONG SEQUENCES ARE TO BE
C      PROCESSED; WHILE THE CALCULATIONS ARE DONE FOR EACH VALUE.
C      ONLY EVERY ISPAC*TH LINE IS ACTUALLY PRINTED OUT.
C
C      THE ROUTINE IMPLEMENTS THE SELF-STARTING CUSUM, AND PRINTS OUT
C      TWO SECTIONS OF PRINTOUT.  IN THE FIRST SECTION ARE SHOWN THE
C      SEQUENCE NUMBER; THE X VALUE; THE CORRESPONDING N(O,1)
C      TRANSFORMED VALUE U; THE RUNNING MEAN AND RUNNING S.D. OF THE
C      X VALUES, AND THE 'CONVENTIONAL' CUSUMS FOR LOCATION AND SCALE.
C      THESE ARE FOLLOWED BY THE FOUR DECISION INTERVAL CUSUMS FOR
C      UPWARD SHIFTS IN (1) LOCATION AND (2) SCALE; AND FOR DOWNWARD
C      SHIFTS IN (3) LOCATION AND (4) SCALE.
C
C      THE SECOND PART OF THE PRINTOUT IS A LINE PRINTER PLOT OF THE
C      DECISION INTERVAL CUSUMS FOR UPWARD AND DOWNWARD SHIFTS IN
C      LOCATION AND SCALE.  IN THIS PLOT, THE LOCATION CUSUMS ARE
C      PLOTTED WITH THE SYMBOL 'L', AND THE SCALE WITH THE SYMBOL
C      'S'.  IF BOTH COINCIDE, THE SYMBOL 'B' IS PLOTTED.
C      THE CUSUMS FOR UPWARD SHIFTS ARE PLOTTED ON THE RIGHT, AND
C      THOSE FOR DOWNWARD SHIFTS ON THE LEFT OF THE PAGE
C
C      IN BOTH THE COMPUTATIONS OF THE RUNNING MEAN AND S.D. AND THE
C      CUSUMS, WINSORISATION IS USED.  OUTLIERS MORE THAN 3 S.D.'S
C      AWAY FROM THE RUNNING MEAN ARE REPLACED BY VALUES AT 3 S.D.'S
C      FROM THE MEAN.  THIS PROCEDURE PROTECTS THE CUSUM FROM THE
C      HARMFUL EFFECTS OF SEVERE OUTLIERS WITHOUT GOING SO FAR AS TO
C      EXCLUDE THEM ENTIRELY.
C      IF WINSORISATION IS NOT DESIRED, SET THE VARIABLE WINSR TO
C      A LARGE VALUE IN THE DATA STATEMENT BELOW.
C
C      THERE IS AN IMPLEMENTATION RESTRICTION PREVENTING USE OF
C      MORE THAN 1250 POINTS.  IF DESIRED, THIS CAN BE INCREASED
C      OR DECREASED BY ADJUSTING THE DIMENSION STATEMENT.
C      AS WRITTEN, THE ROUTINE PLACES FOUR LARGE WORK ARRAYS
C      IN THE NAMED COMMON BLOCK /WORK/.  IF THIS FEATURE IS NOT
C      WANTED, THEN THE COMMON STATEMENT MAY BE DELETED
C
C      DOUGLAS M HAWKINS
C
C      *****
C
C      DIMENSION LP(1250),SP(1250),LM(1250),SM(1250),X(N),
1 LINEOT(101)
C      REAL LP, LM
C      COMMON /WORK/ LP, SP, LM, SM
C      DATA IBLA/1H /,ISS/1HS/,ILL/1HL/,IDBL/1HB/,IVER/1H' /,WINSR/3./
1, IPRTER/6/
C      ISPAC = MAXO(ISPAC,1)
C      WIN2 = WINSR * WINSR
C      CS = 0
C      CM = 0
C      CSL = 0
C      CSS = 0
C      FI = 0
C      DIPL = 0
C      DIML = 0
C      DIPS = 0
C      DIMS = 0
C      PMAXL = 0

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      PMINL = 0
      WRITE(IPRTER,100)
100  FORMAT(45H1SELF STARTING CUSUMS FOR LOCATION AND SCALE /
      1 3X,2HNO,T11,1HX,7X,1HU,10X,7HRUNNING,5X,19HCONVENTIONAL CUSUMS
      1 3X,24HDECISION INTERVAL CUSUMS/
      2 T27,4HMEAN,5X,4HS.D.,3X,8HLOCATION,4X,5HSCALE,
      3 T65,2HLP,5X,2HLM,5X,2HSP,5X,2HSM/)
      LOCST = 0
      DO 1 I = 1, N
      XNEW = X(I)
      FI = FI + 1
      FIR = (FI - 1.) / FI
      DEV = XNEW - CM
      U = 0
      IF (FI .LT. 2.5 .OR. CS .LE. 0.) GO TO 2
      DEGRF = FI - 2
      FMULT = (8 * DEGRF + 1) / (8 * DEGRF + 3)
      SONUM = DEV * DEV * FIR
      SIGD = SIGN(1., DEV)
      U = FMULT * SQRT(DEGRF * ALOG(1 + SONUM / CS)) * SIGD
      IF (ABS(U) .LT. WINSR) GO TO 3
      EXTER = (WINSR / FMULT) ** 2 / DEGRF
      DEV = SIGD * SQRT(CS * (EXP(EXTER) - 1) / FIR)
      U = WINSR * SIGD
3    CONTINUE
      CSL = CSL + U
      SCT = (SQRT(ABS(U)) - .822) / .349
      CSS = CSS + SCT
      DIPL = AMAX1(0., DIPL + U - AK)
      DIML = AMIN1(0., DIML + U + AK)
      DIPS = AMAX1(0., DIPS + SCT - AK)
      DIMS = AMIN1(0., DIMS + SCT + AK)
2    CONTINUE
      CM = CM + DEV / FI
      CS = CS + DEV ** 2 * FIR
      IF(MOD(I,ISPAC) .NE. 0) GO TO 1
      LOCST = LOCST + 1
      PMAXL = AMAX1(PMAXL, DIPL, DIPS)
      PMINL = AMIN1(PMINL, DIML, DIMS)
      LP(LOCST) = DIPL
      LM(LOCST) = DIML
      SP(LOCST) = DIPS
      SM(LOCST) = DIMS
      UNSD = SQRT(CS / FI)
      WRITE(IPRTER,102) FI,XNEW,U,CM,UNSD,CSL,CSS,DIPL,DIML,DIPS,DIMS
102  FORMAT(F6.0,F9.3,F6.2,2F9.3,2F9.1,2X,4F7.2)
1    CONTINUE
20   CDNTINUE
      IUP = PMAXL / 5. + 1.
      IDOWN = -PMINL / 5. + 1
      IF (MOD(IUP+IDOWN,2) .EQ. 0) GO TO 11
      IF (IUP .LT. IDOWN) IUP = IUP + 1
      IF (IDOWN .LT. IUP) IDOWN = IDOWN + 1
11   PMAXL = 5. * IUP
      PMINL = -5. * IDOWN
      RANGE = PMAXL - PMINL
      LOW = PMINL - 1.E-5
      ITOP = PMAXL + 1.E-5
      SCALF = 100. / (PMAXL - PMINL)
      IORG = 1.5 - SCALF * PMINL
      INTER = .1 * RANGE + 1.E-5
      WRITE(IPRTER,111) (I, I = LOW, ITOP, INTER)
111  FORMAT(30H1CUSUMS FOR LOCATION AND SCALE/T21,11110)
      WRITE(IPRTER,122)
122  FORMAT(T30,10(10H+-----),1H+)
      DO 21 J = 1, LOCST
      IXL = 1.5 + SCALF * (LP(J) - PMINL)
      IYL = 1.5 + SCALF * (SP(J) - PMINL)
      IXN = 1.5 + SCALF * (LM(J) - PMINL)
      IYN = 1.5 + SCALF * (SM(J) - PMINL)
      DO 25 I = 1, 101
      LINEOT(I) = IBLA
      LINEOT(IXL) = ILL
      LINEOT(IYL) = ISS
      IF (IXL .EQ. IYL) LINEOT(IXL) = IDBL
      LINEOT(IXN) = ILL
      LINEOT(IYN) = ISS
      IF (IXN .EQ. IYN) LINEOT(IXN) = IDBL
      LINEOT(IORG) = IVER
      NUM = J * ISPAC
      WRITE(IPRTER,112) NUM,LINEOT
112  FORMAT(I5,T30,101A1)
21   CONTINUE
30   CONTINUE
      END

```